Computer Vision Programming Assignment 2

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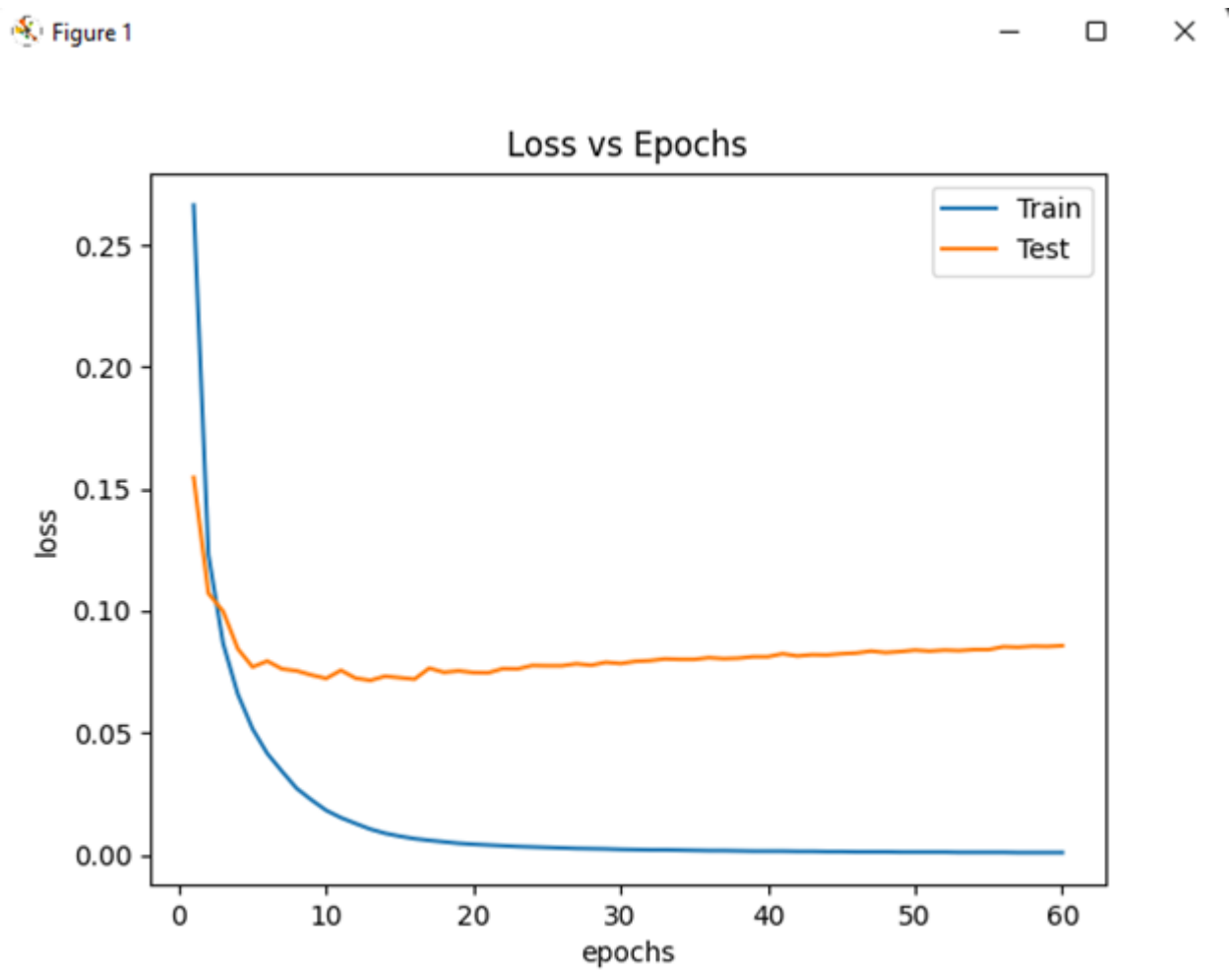
Question: Convolution Neural Network for Digit Classification

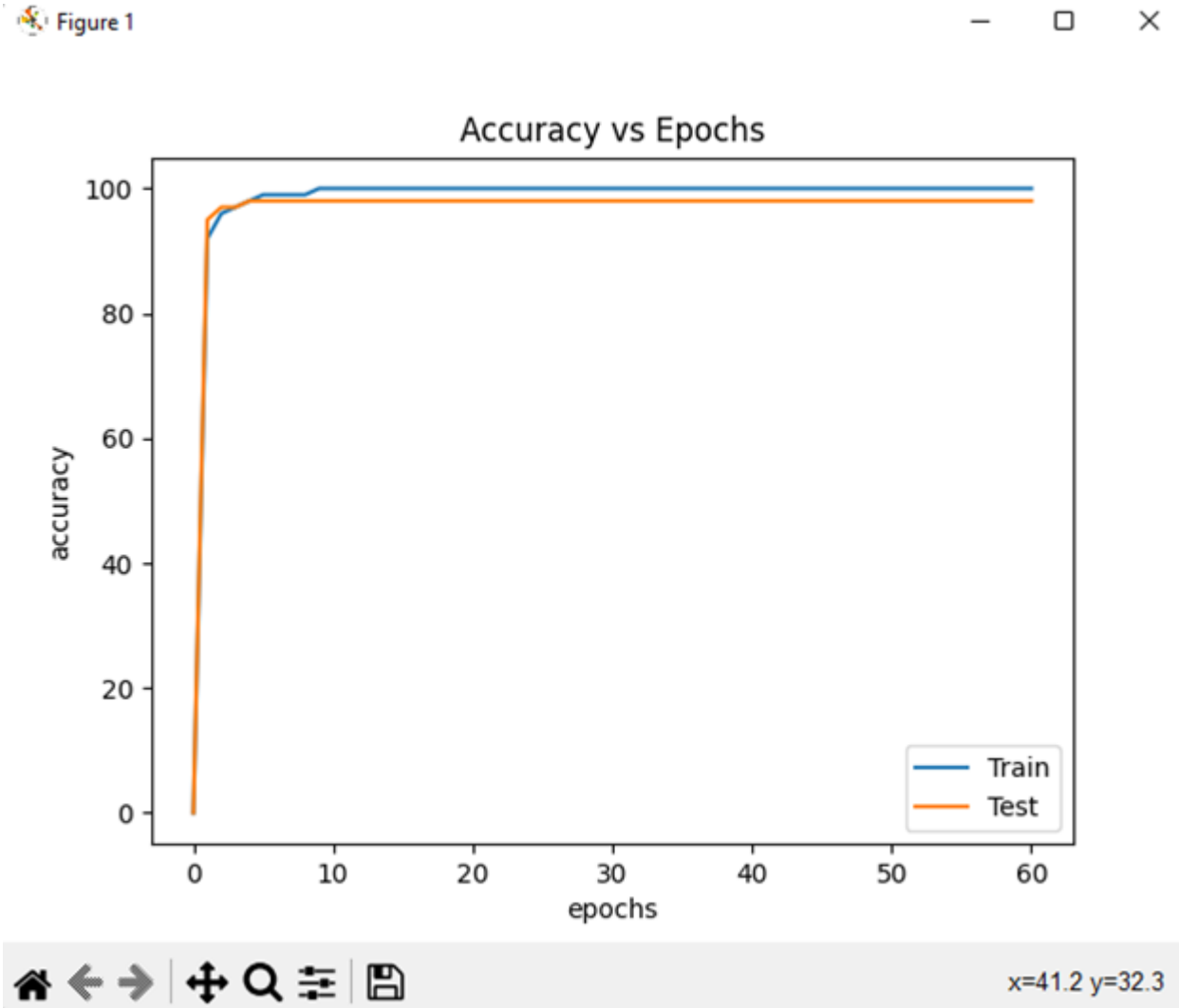
Steps:

Step 1: To create a fully connected hidden layer using sigmoid activation function with 100 neurons. To train it with learning rate of 0.1, 60 epoch, mini batch size of 10 and no regularization. For this model, I got maximum accuracy of 98%.

Graphs of loss vs epoch

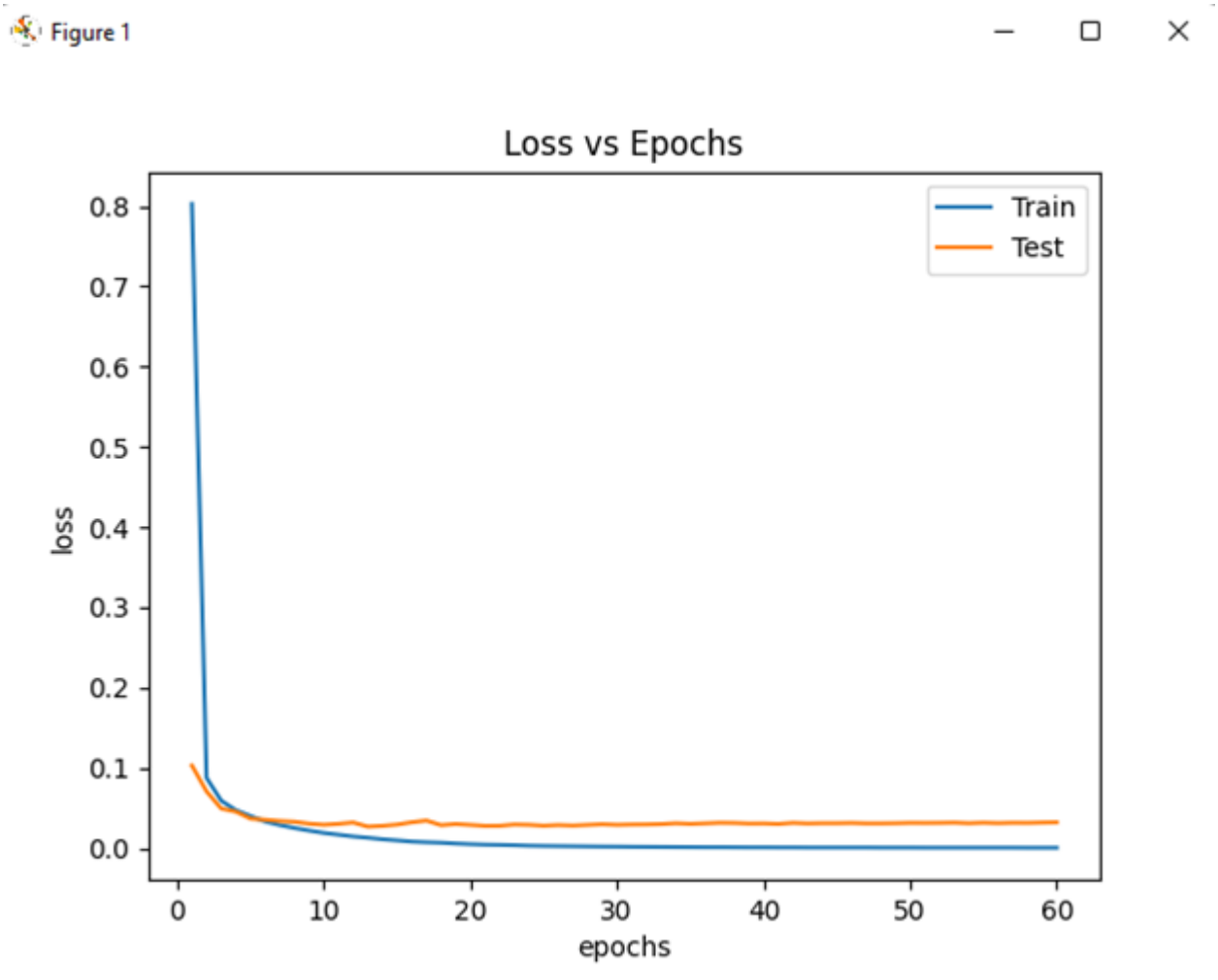


Graph of accuracy vs epoch

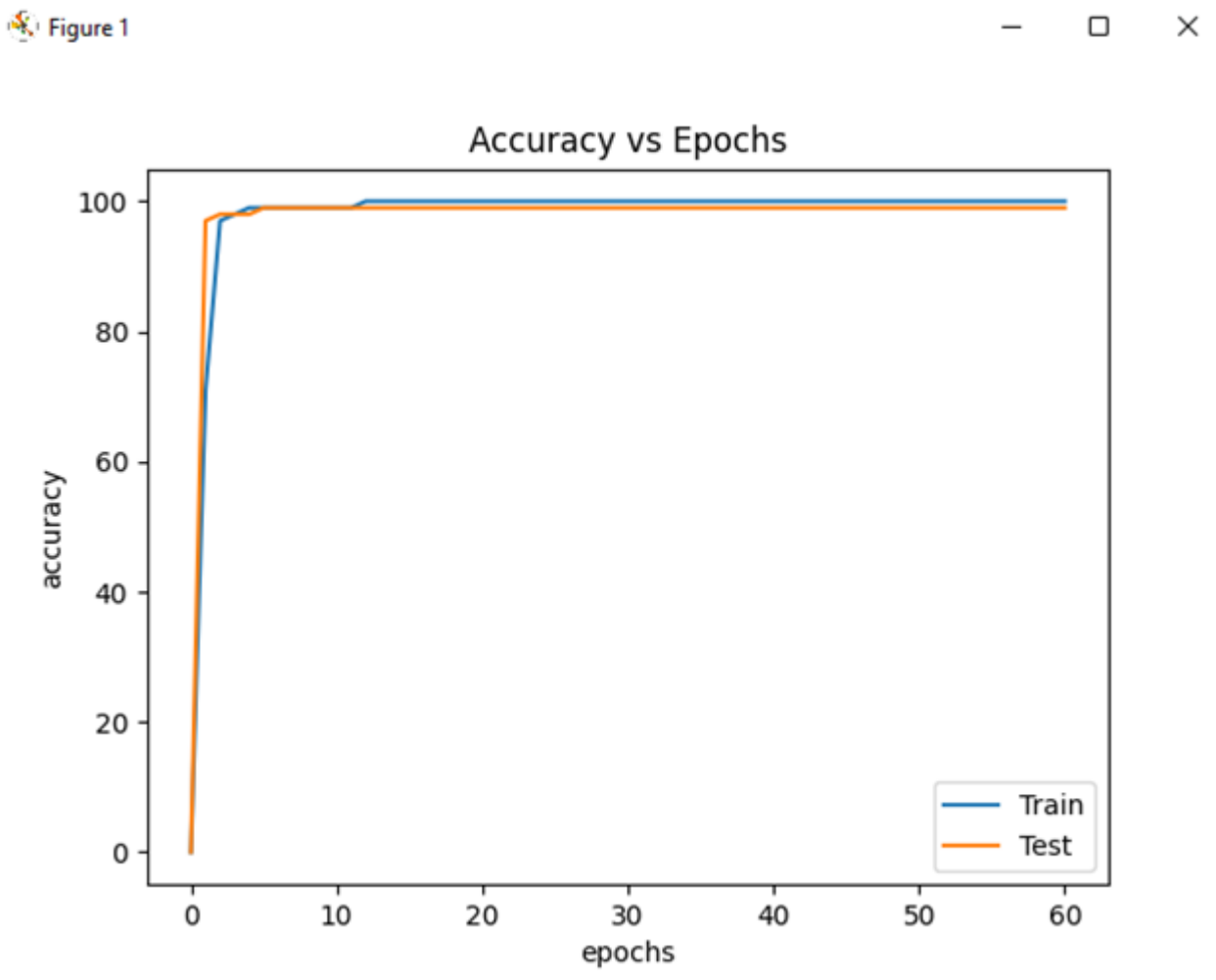


Step 2: To insert two convolutional networks from step 1 and put pooling layer to each convolution network with pool over 2*2 regions, 40 kernels, stride =1, kernel size = 5*5. I got maximum accuracy of 99% for this model.

Graph of loss vs epoch

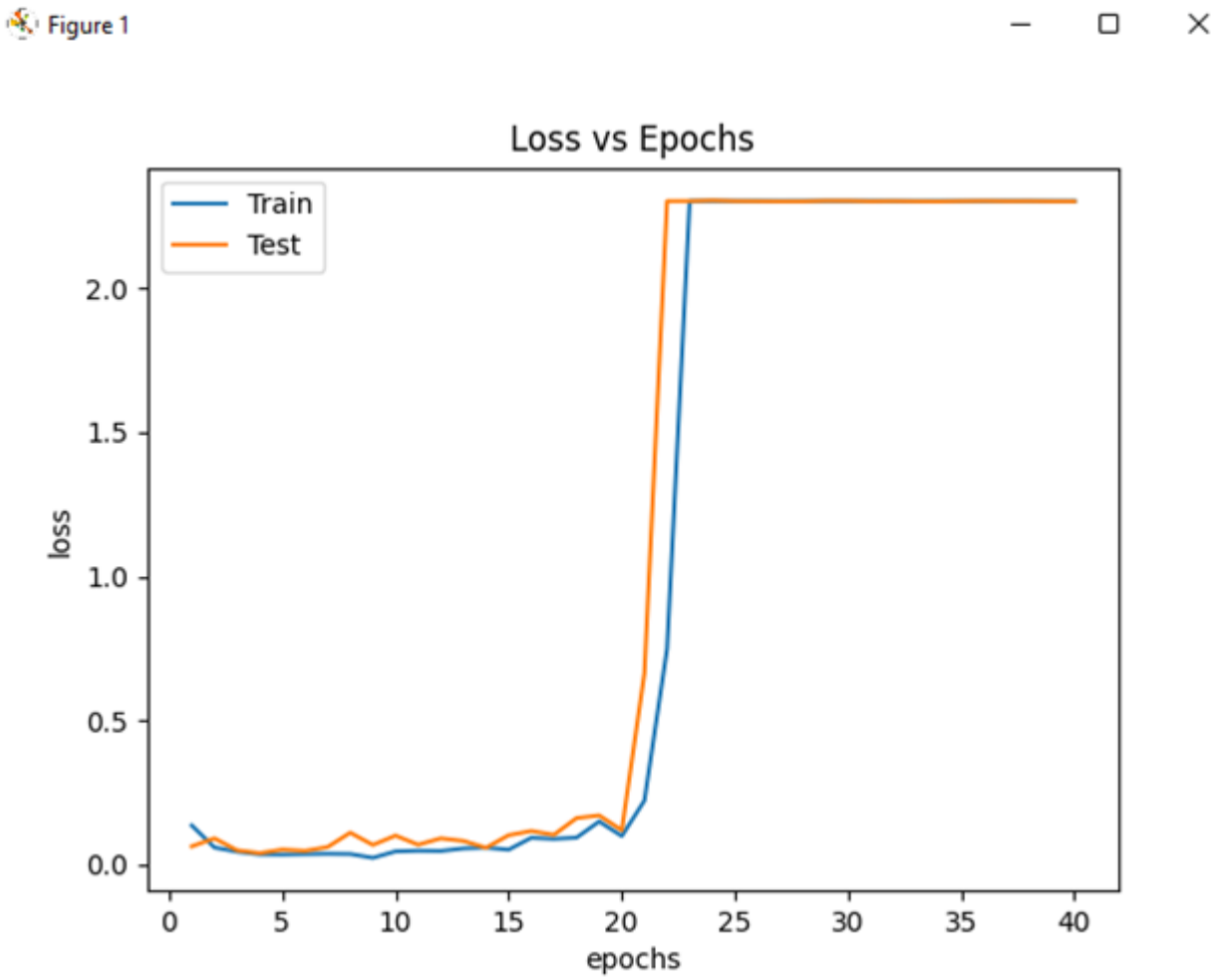


Graph of accuracy vs epoch

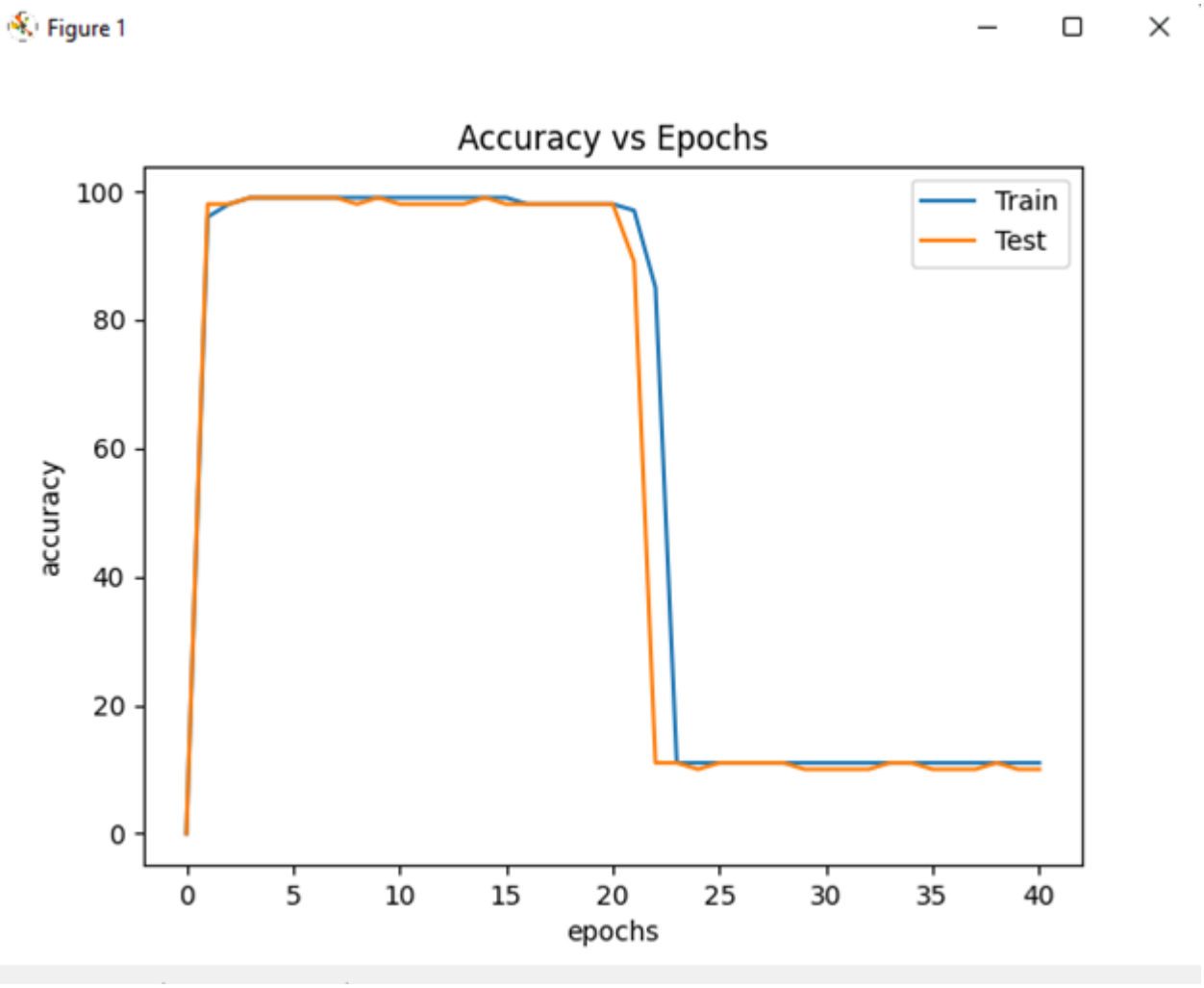


Step 3: Replace sigmoid activation function with ReLu and train the model with new learning rate of 0.03. I got accuracy of 99.03% for this model.

Graph of loss vs epoch

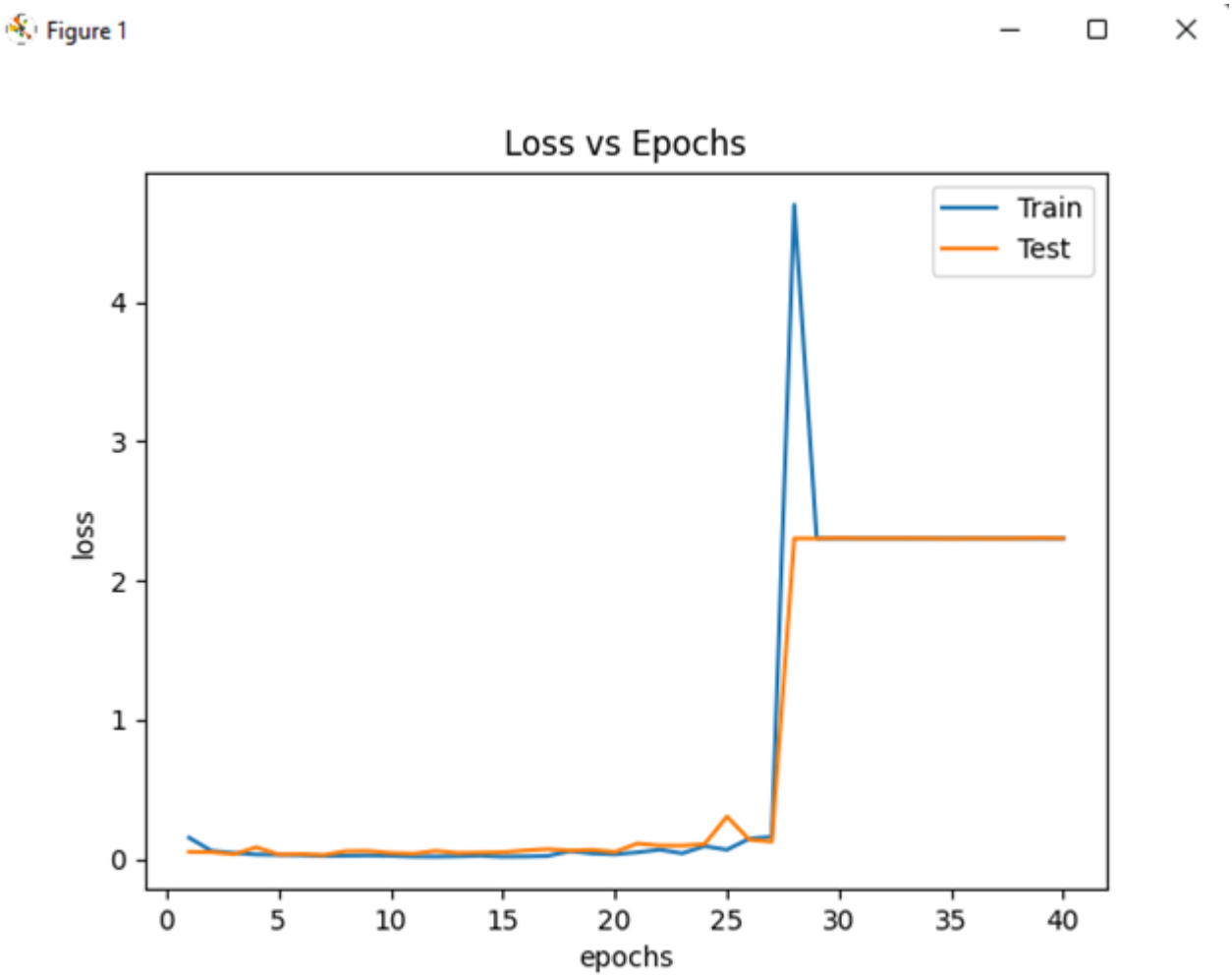


Graph of accuracy vs epoch

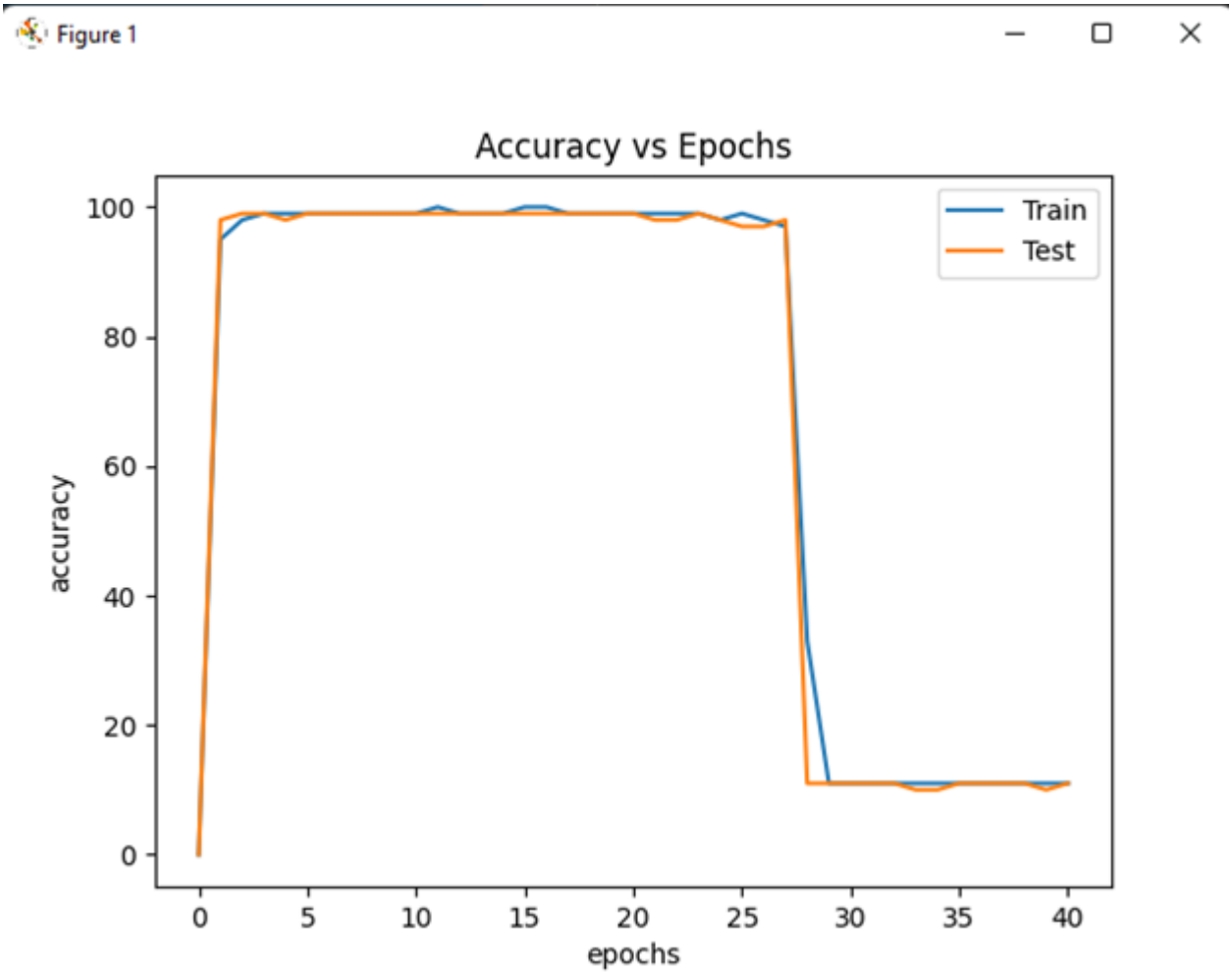


Step 4: To add another fully connected network to this network with 100 neurons. I got accuracy of 99.15%.

Graph of loss vs epoch

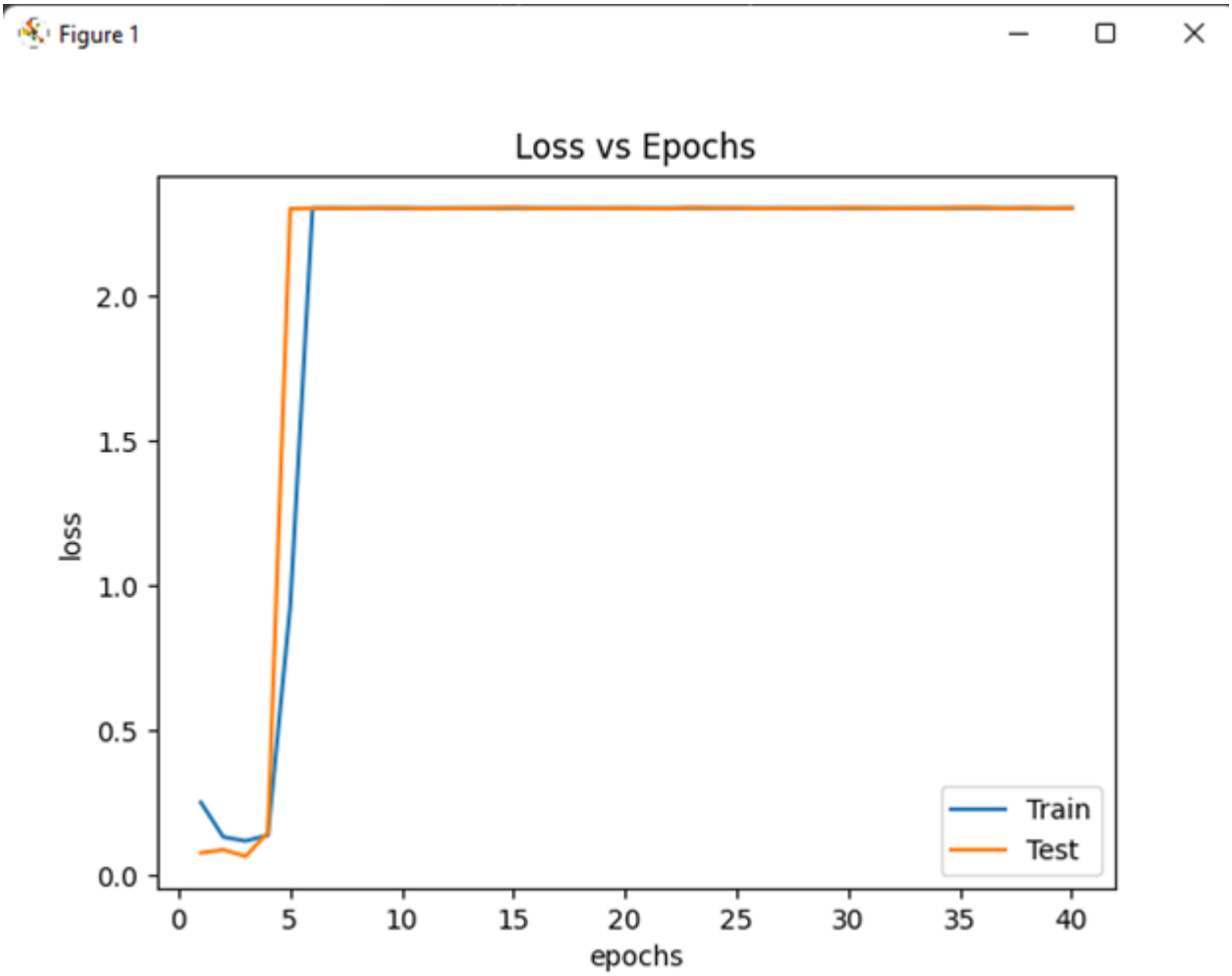


Graph of accuracy vs epoch



Step 5: To change the number of neurons to 1000 in fully connected layer. For regularization use dropout with rate 0.5. To train the system with 40 epoch. Accuracy is 98.51%.

Graph of loss vs epoch



Graph of accuracy vs epoch

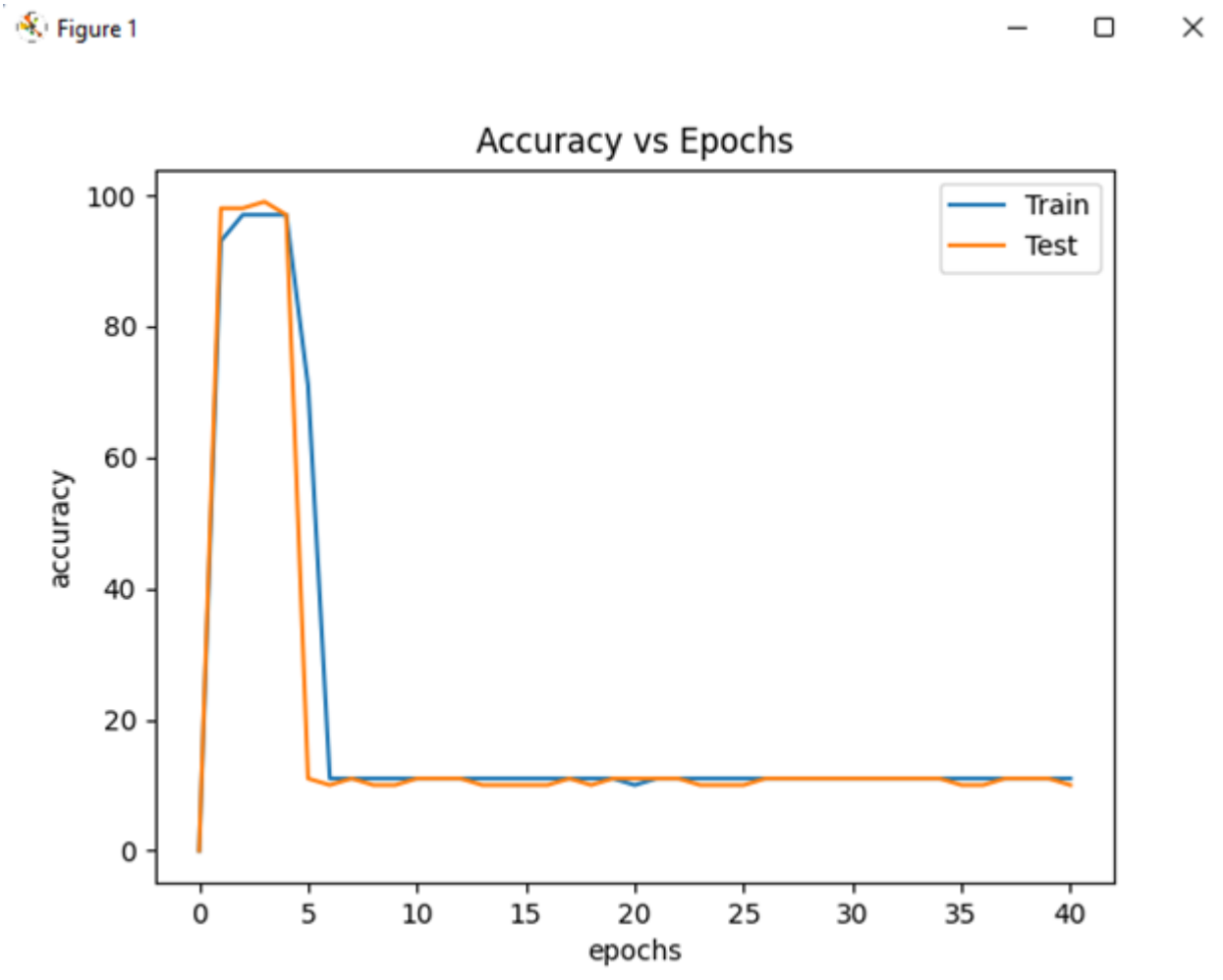


Figure 1
Accuracy vs Epochs
Train
Test
accuracy
epochs
100
80
60
40
20
0
0
5
10
15
20
25
30
35
40